

Introduction

A registered report separates *what a study commits to* from *what the data later show*. Hypotheses, outcomes, exclusion rules, and the analysis plan are locked and time-stamped before results are observed; only then is the analysis run and interpreted. This ordering is the structural defence against the degrees-of-freedom problems that inflate false-positive rates in flexible analyses (Simmons et al. 2011) and motivates the broader preregistration and research-transparency movement (Nosek et al. 2018; Munafò et al. 2017; Chambers 2013).

This template operationalises that discipline as tested code rather than prose convention. The `registered_report` package freezes a registration payload under a content hash, validates that the required preregistration sections are present and well formed, and — after execution — compares the analysis that was actually run against the plan that was locked. Any departure must appear in a **deviation ledger** with a rationale before it can be reported, and confirmatory claims are kept mechanically distinct from exploratory ones.

To make the workflow concrete and reproducible, the template ships a **deterministic demonstration study**. It contains no real empirical data: a seeded generator synthesises a two-group dataset, and the registered label-permutation test is executed against it. Because both the data and the permutation schedule are seeded, every figure and every number in this document is byte-stable and is regenerated on demand from `scripts/generate_figures.py`. Readers forking this template should replace the demonstration fixture in `data/example_registration.json` and the synthetic data generator with their own registration and real data collection, keeping the same lock-validate-execute-compare structure.

The remainder of this report states the preregistered hypotheses (Section `sec:hypotheses`), describes the frozen methods and analysis plan (Section `sec:methods`), reports the confirmatory outcome bound to the executed analysis (Section `sec:results`), records deviations (Section `sec:deviations`), and discusses scope and boundaries (Section `sec:discussion`).